

Tutorial 4 - Econometric Theory

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Where we left?

There are many reasons for which **naive TWFE** is a **BAD idea in non-standard settings**

It is the 'simplest' estimator for a staggered setting (that's why everybody use it)

Used when:

- Differential Timing in treatment adoption
- Binary Treatment (that stays on and never turns off!!!! **Absorbing**)
- Unrestricted heterogenous treatment effects (i.e. different groups are allowed to respond differently to the treatment)

What it estimates?

$$ATT(g, t) = \mathbb{E}[Y_t(g) - Y_t(\infty) | G = g], t \geq g$$

You get one of these for each group, then you can aggregate them as you prefer

In practice, identified objects are:

$$ATT_{g,t}^{nev} = \mathbb{E}[Y_t - Y_{g-1} | G_i = g] - \mathbb{E}[Y_t - Y_{g-1} | G_i = \infty]$$

$$ATT_{g,t}^{ny} = \mathbb{E}[Y_t - Y_{g-1} | G_i = g] - \mathbb{E}[Y_t - Y_{g-1} | G_i \neq g, W_t = 0]$$

Pay attention to the reference period!!! It is $g-1$ always (long-difference, as in the dynamic TWFE).

You will see a discussion about this in a moment.

Aggregation Methods

- Standard Weighted Aggregation

$$\theta^{agg} = \sum_{g=2}^T \sum_{t=g}^T \mathbb{1}\{g \leq t\} ATT(g, t) \cdot w_{g,t}$$

where:

$$w_{g,t} = Pr(G = g | G \leq T) \cdot \frac{\mathbb{1}\{g \leq t\}}{T - g + 1}$$

- $Pr(G = g | G \leq T)$ gives more weight to larger treatment cohorts among ever-treated
- $\frac{\mathbb{1}\{g \leq t\}}{T - g + 1}$ gives equal weight across all post-treatment periods available for each cohort g
- Pre-treatment cells get weight 0 because of $\mathbb{1}\{g \leq t\}$

Aggregation Methods

- ATT for units of group g :

$$\theta(g) = \frac{1}{T - g + 1} \sum_{t=2}^T \mathbb{1}\{g \leq t\} ATT(g, t)$$

- ATT in time t for groups which had been treated by time t :

$$\theta(t) = \sum_{g=1}^t \mathbb{1}\{g \leq t\} ATT(g, t) P(G = g | G \leq t, G \neq \infty)$$

- ATT for the groups of units that have been exposed to treatment for exactly ϵ periods:

$$\theta(\epsilon) = \sum_{t=g}^T \mathbb{1}\{g + \epsilon \leq t\} ATT(g, g + \epsilon) P(G = g | G + \epsilon \leq T, G \neq \infty)$$

Assumptions

Assumptions:

- No Anticipation
- Irreversible Treatment (Absorbing)
- (Conditional) Parallel Trends (remind me to talk about covariates problems in CS if I forget)
- Additional when adding covariates:
 - Common Support / Correct regression specification

PTA framing:

- Relative to Never-Treated:

$$\mathbb{E}[Y_t(\infty) - Y_{g-1}(\infty) | G_i = g] = \mathbb{E}[Y_t(\infty) - Y_{g-1}(\infty) | G_i = \infty]$$

- Relative to Not-Yet-Treated:

$$\mathbb{E}[Y_t(\infty) - Y_{g-1}(\infty) | G_i = g] = \mathbb{E}[Y_t(\infty) - Y_{g-1}(\infty) | G_i \neq g, W_t = 0]$$

Example

Truth					CS estimates				
Year	ATT(1986,t)	ATT(1992,t)	ATT(1998,t)	ATT(2004,t)	Year	ATT(1986,t)	ATT(1992,t)	ATT(1998,t)	ATT(2004,t)
1980	0	0	0	0	1981	-0.0548	0.0191	0.0578	0
1986	10	0	0	0	1986	10.0258	-0.0128	-0.0382	0
1987	20	0	0	0	1987	20.0439	0.0349	-0.0105	0
1988	30	0	0	0	1988	30.0028	-0.0516	-0.0055	0
1989	40	0	0	0	1989	40.0201	0.0257	0.0313	0
1990	50	0	0	0	1990	50.0249	0.0285	-0.0284	0
1991	60	0	0	0	1991	60.0172	-0.0395	0.0335	0
1992	70	8	0	0	1992	69.9961	8.013	0	0
1993	80	16	0	0	1993	80.0155	16.0117	0.0105	0
1994	90	24	0	0	1994	89.9912	24.0149	0.0185	0
1995	100	32	0	0	1995	99.9757	32.0219	-0.0505	0
1996	110	40	0	0	1996	110.0465	40.0186	0.0344	0
1997	120	48	0	0	1997	120.0222	48.0338	-0.0101	0
1998	130	56	6	0	1998	129.9164	56.0051	6.027	0
1999	140	64	12	0	1999	139.9235	63.9884	11.969	0
2000	150	72	18	0	2000	150.0087	71.9924	18.0152	0
2001	160	80	24	0	2001	159.9702	80.0152	23.9656	0
2002	170	88	30	0	2002	169.9857	88.0745	29.9757	0
2003	180	96	36	0	2003	179.981	96.0161	36.013	0
2004	190	104	42	4	2004				
2005	200	112	48	8	2005				
2006	210	120	54	12	2006				
2007	220	128	60	16	2007				
2008	230	136	66	20	2008				
2009	240	144	72	24	2009				
ATT	82				Total ATT	n/a			
Feasible ATT	68.3333333				Feasible ATT	68.33718056			

Discussion of long-differences

When estimating leads and lags in TWFE dynamic regression, you were to drop one year dummy variable, hence interpreting each estimated coefficient as relative to a universal baseline (usually $t-1$)

In CS (as in dCDH), there are two ways to calculate the pre-treatment coefficients once you have obtained the $ATT(g,t)$ parameter estimates:

- **Long-difference:** universal baseline fixed in $t-1$

$$\begin{aligned}\hat{\delta}_{t-3} = & (E[Y|D = 1, t - 3] - E[Y|D = 1, t - 1]) \\ & - (E[Y|D = 0, t - 3] - E[Y|D = 0, t - 1])\end{aligned}$$

- **Short-difference:** 'rolling' method in which a new baseline is used for each 2X2 comparison

$$\begin{aligned}\hat{\delta}_{t-3} = & (E[Y|D = 1, t - 3] - E[Y|D = 1, t - 2]) \\ & - (E[Y|D = 0, t - 3] - E[Y|D = 0, t - 2])\end{aligned}$$

Discussion on long-differences

In practice, the two methodology can show parallel pre-trends in one case but not in the other.

Why?

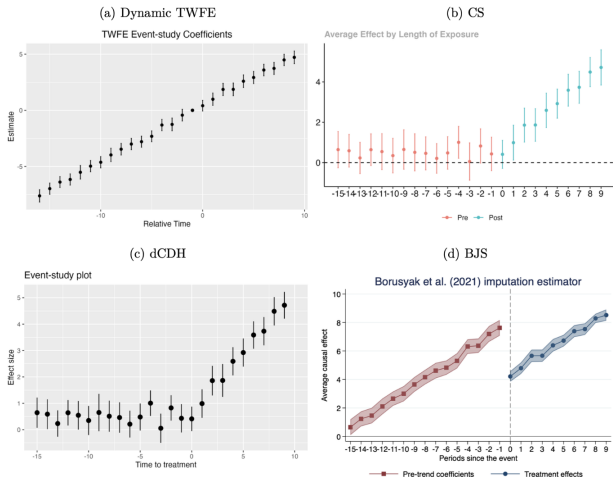
Weird because, in principle, they are representing the same object. And you can recover one from the other.

The **answer** is: the event study is a **falsification exercise**, it is not telling you any truth about pre-trends. It just helps in justifying the PT assumption.

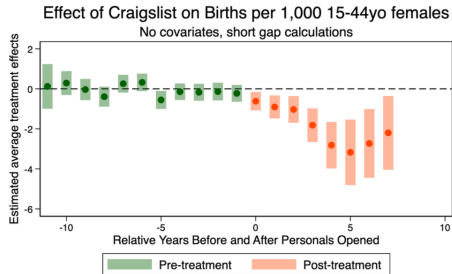
So, what one should use? Long differences → because A) for each ATT you are using $t-1$ as baseline, hence keep everything consistent B) Easier to spot pre-trends (and falsify them, check the example later)

Discussion on long-differences (Roth 2024 paper)

Figure 1: Comparison of event-study plots in a non-staggered setting

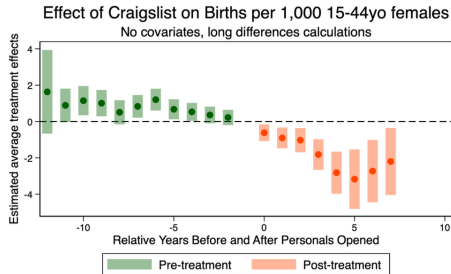


Discussion on long-differences (Scott Working Paper)



Note: Uses the 2008–2010 eventually treated and the not-yet-treated counties as controls, but no covariates. Circles are $ATT(g,t)$ estimates by relative event time. All groups and bands are 95% uniform confidence intervals. Mean birth rate was approximately 0.062 in 2000.

Figure: Short Gap



Note: Uses the 2008–2010 eventually treated and the not-yet-treated counties as controls, but no covariates. Circles are $ATT(g,t)$ estimates by relative event time. All groups and bands are 95% uniform confidence intervals. Mean birth rate was approximately 0.062 in 2000.

Figure: Long Difference

Last Reminders

Long vs Short Differences:

- Better to use long differences. To do so, select the option `long2` (both in Stata and R)
- Easy to spot what is being used in a paper: if long difference, no coefficient estimated in $t-1$. If you see a coefficient there, and the guy presenting did not say anything about it *start worrying*. It was ok not knowing it until few years ago, not now.

Last Reminders

Covariates:

- In CSDiD you can insert covariates, with the three usual methods (IPW, RA, DR). DR is recommended.
- If you do it, you are "very likely" subject to additional assumptions (common support, correct regression specification).
- Notice that, for IPW (hence DR as well), you are *estimating treatment probabilities* with logit/probit. It is very likely that you will obtain some units with very high (≈ 1) or very low (≈ 0) predicted probabilities, hence with very high weights. This fucks up everything.
- Solution: trim the sample, dropping units with predicted probabilities out of the 5% confidence interval (treat them as outliers).
- This is done *automatically* in R, **not in Stata**. And you cannot do it with the CS package AT ALL!! You cannot even see the assigned weights (they get dropped after the estimation). You need to do all the estimation by hand, in order to solve the problem.

Synthetic Control Methods

What does it mean to use a synthetic control method?

- T and C groups are not good enough for making inference (i.e. no PTA, few (even 1) treated units, ...)
- Idea: create a synthetic control group, as a weighted average of the available controls, with weights chosen to make the synthetic control as similar as possible to the treated unit in the pre-treatment period.
- In principle a very smart idea, a "revolution" for policy evaluation

Synthetic Control - Setup

Suppose we observe $J + 1$ units in periods $1, 2, \dots, T$

- Unit 1 is exposed to the intervention during periods $T_0 + 1, \dots, T$
- The remaining J units form an untreated “donor pool”
- Define the treatment effect for the treated unit at time $t > T_0$:

$$\theta_{1t} = Y_{1t}(1) - Y_{1t}(0)$$

- we observe $Y_{1t}(1)$ but need to estimate $Y_{1t}(0)$
using the donor pool

Synthetic Control - Setup

- Let $W = (w_2, \dots, w_{J+1})'$ with $w_j \geq 0$ and $\sum_j w_j = 1$
 - Each W defines a potential synthetic control
- Let X_1 be a $(k \times 1)$ vector of pre-intervention characteristics for the treated unit, and X_0 a $(k \times J)$ matrix for the donor pool
- Choose W^* to minimize the distance:

$$W^* = \arg \min_W \|X_1 - X_0 W\|_V$$

where $\|x\|_V = \sqrt{x' V x}$ and V is a $(k \times k)$ positive semidefinite matrix of predictor importance weights

Synthetic Control - Setup

- V determines the relative importance of each predictor when computing the distance between treated and synthetic units
 - E.g. if $V = I$: standard Euclidean distance (all predictors matter equally)
 - if $V \neq I$: some predictors matter more than others (e.g. **lagged outcome**).
- V^* directly shapes W^* : different $V \Rightarrow$ different synthetic control
- Default approach: choose V to minimize pre-treatment Mean Squared Prediction Error (MSPE):

$$V^* = \arg \min_V \sum_{t=1}^{T_0} \left(Y_{1t} - \sum_{j=2}^{J+1} w_j^*(V) Y_{jt} \right)^2$$

Synth Estimator

So, the **synthetic control estimator** is:

$$\hat{\beta} = Y_{1t} - \sum_{j=2}^{J+1} w_j^* Y_{jt}$$

that is, the difference between the treated unit and the synthetic control in the post-treatment periods.

Requirements / Assumptions

SC requires the synthetic control to lie inside the **Convex Hull** of the donor pool, that is:

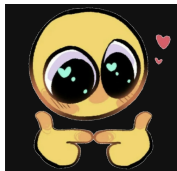
$$\begin{cases} w_j \geq 0 \\ \sum_j w_j = 1 \end{cases}$$

- Avoids extrapolations
- but this make pre-treatment fit usually worse

Examples

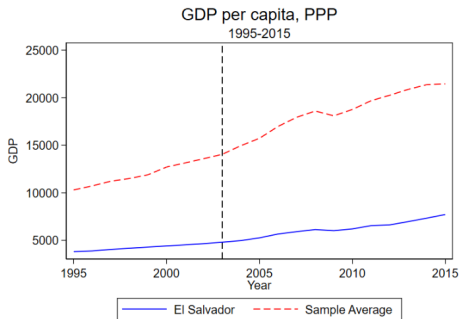
Following examples are taken from my master thesis.

Most of this stuff ex-post is terrible, but it is just easier for me to make examples with stuff I know. Assume that everything is correct (spoiler: it is not) and keep it as a reference.

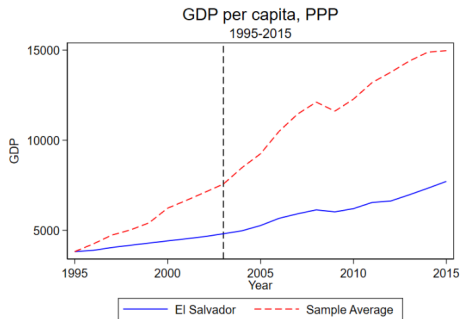


Please be kind :)

Examples



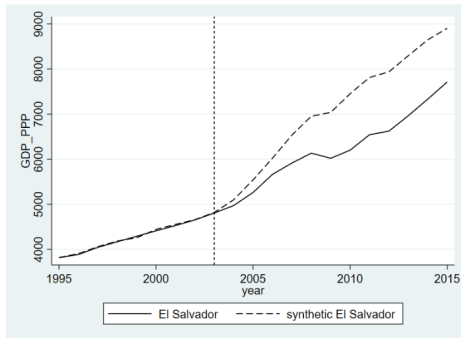
(a) El Salvador against the Sample Average



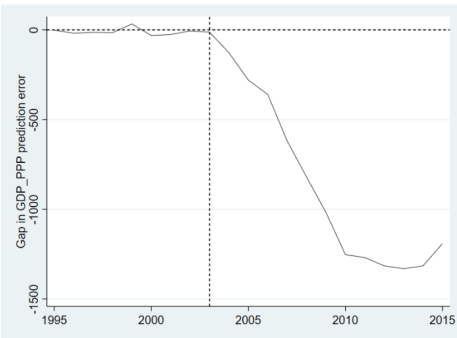
(b) Same as a, but equating the starting level

Figure 3.2: GDP comparisons

Examples



(a) Synthetic El Salvador



(b) Gap between treated and counterfactual

Examples

	Treated	Synthetic	Sample Average
GDP_PPP(1998)	4167.94	4183.48	11501.18
GDP_PPP(2000)	4410.68	4442.13	12706.57
GDP_PPP(2003)	4805.72	4818.14	14044.83
Mean_HCest	8.63	7.83	12.05
IR	17.58	18.01	22.41
PD	285.19	122.41	172.59
Public_Exp	12.29	12.12	15.74
Trade	2857.56	2685.51	2198.46
VAagr	2971.22	2879.26	2392.04
VAind	2396.11	2379.21	2304.89
VAmam	2091.67	2398.00	2174.98
VAsr	57.33	55.99	50.60

Table 5.1: Pre-Treatment Period Matching

Examples

Country	Weight	Region	Income Group	IH level
Bulgaria	.061	Europe & Central Asia	Upper middle income	4.30
Dominican Republic	.175	Latin America & Caribbean	Upper middle income	14.14
Gambia	.178	Sub-Saharan Africa	Low income	/
Haiti	.159	Latin America & Caribbean	Lower middle income	/
Jamaica	.042	Latin America & Caribbean	Upper middle income	36.47
Kazakhstan	.028	Europe & Central Asia	Upper middle income	14.95
Peru	.117	Latin America & Caribbean	Upper middle income	/
Uruguay	.072	Latin America & Caribbean	High income	6.66
Vanuatu	.169	East Asia & Pacific	Lower middle income	/

Notes: Along with the countries' weights, the Region and Income Group column reports the geographical location and the income level, respectively. The last column shows the Intentional Homicides mean level, averaged in the pre-treatment period. The "/" symbol indicates missing data for IH level.

Table 5.2: Country Weights

Inference in Synth Control

Two basic things one should always do:

- **"In-time" placebo** → applying the synthetic control method to the same treated unit but in periods of time in which the intervention did not take place.
- **"In-space" placebo** → applying the synthetic control to each of the units in the donor pool (exclude actual treated unit from the sample). Then, compare the distribution of the placebo effects with the estimated effect for the treated unit.

Inference in Synth Control

The procedure is conducted as follows: iteratively apply the synthetic control method to each country in the donor pool to obtain a distribution of placebo effects. Then calculate the RMSPE (Root Mean Squared Prediction Error) for each placebo for the post-treatment period as:

$$RMSPE_{post} = \left(\frac{1}{T - T_0} \sum_{t=T_0+1}^T \left(Y_{1t} - \sum_{j=2}^{J+1} w_j^* Y_{jt} \right)^2 \right)^{\frac{1}{2}} \quad (4.4)$$

Calculate the RMSPE for each placebo, but for the pre-treatment period as:

$$RMSPE_{pre} = \left(\frac{1}{T_0} \sum_{t=1}^{T_0} \left(Y_{1t} - \sum_{j=2}^{J+1} w_j^* Y_{jt} \right)^2 \right)^{\frac{1}{2}} \quad (4.5)$$

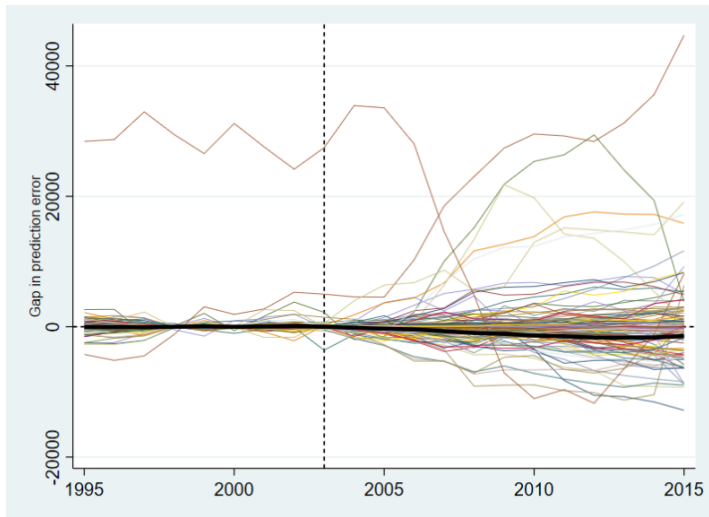
Compute the ratio of the post- to pre-treatment RMSPE and sort it in descending order, assigning a lower value (RANK) the higher is this ratio³. The "p-value" will be calculated as:

$$p = \frac{RANK}{TOTAL}, \text{ where } TOTAL \text{ is the total number of countries in the sample} \quad (4.6)$$

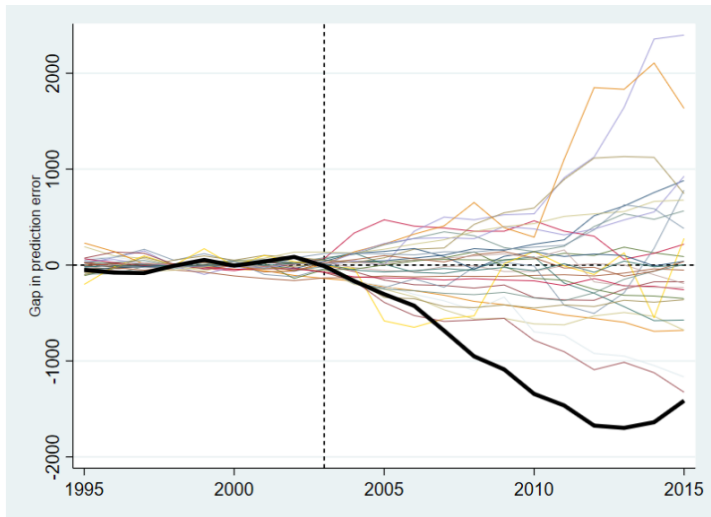
The value of p will reflect the relative position of the unit effect in the created distribution.

³RANK values range from 1 to the number of countries in the sample.

Examples



Examples



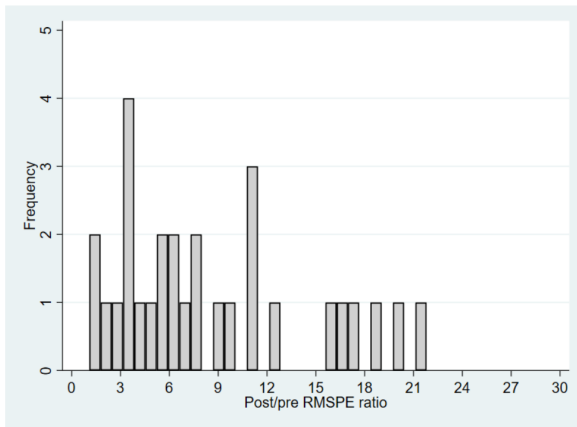
Examples

Country	RMSPE_Pre	RMSPE_Post	Ratio
Bangladesh	22.721	426.450	18.769
Benin	12.011	245.020	20.400
Burkina Faso	48.138	92.179	1.915
Burundi	63.901	470.012	7.355
Cambodia	58.845	345.763	5.876
Chad	34.634	340.651	9.836
Colombia	72.541	1199.530	16.536
Comoros	81.849	507.303	6.198
Congo, Dem. Rep.	97.078	105.006	1.082
Cote d'Ivoire	94.609	331.870	3.508
El Salvador	54.860	1196.211	21.805
Gambia	25.542	290.637	11.379
Ghana	64.192	709.089	11.046
Guinea	43.288	73.692	1.702
Guyana	100.482	384.845	3.830
Haiti	44.197	699.255	15.821
India	30.301	106.135	3.503
Kyrgyz Republic	74.526	821.509	11.023
Madagascar	51.535	195.827	3.800
Mongolia	95.394	1158.422	12.144
Nepal	27.838	256.803	9.225
Niger	41.979	176.636	4.208
Pakistan	25.244	430.978	17.073
Rwanda	83.436	445.597	5.341
Senegal	31.442	79.513	2.529
Sierra Leone	51.667	316.394	6.124
Tanzania	9.940	47.973	4.826
Togo	49.400	385.763	7.809
Uganda	21.129	143.975	6.814

Notes: Only countries with a pre-treatment RMSPE lower than two times the pre-treatment RMSPE of El Salvador are shown.

Table 5.3: RMSPE Statistics by Country

Examples



Notes: El Salvador is represented by the rightest bar, above a value of 21

Figure 5.4: Gap intensities for selected countries

Examples

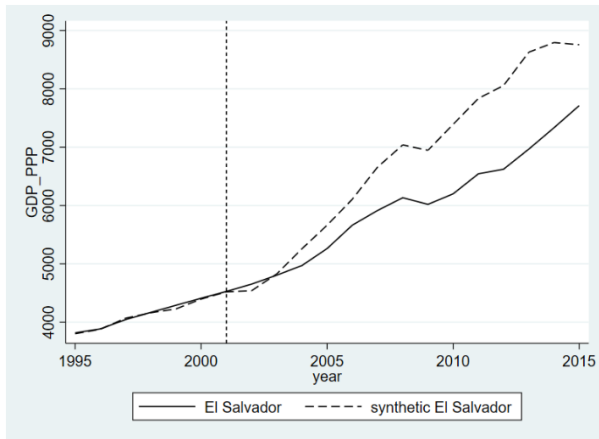
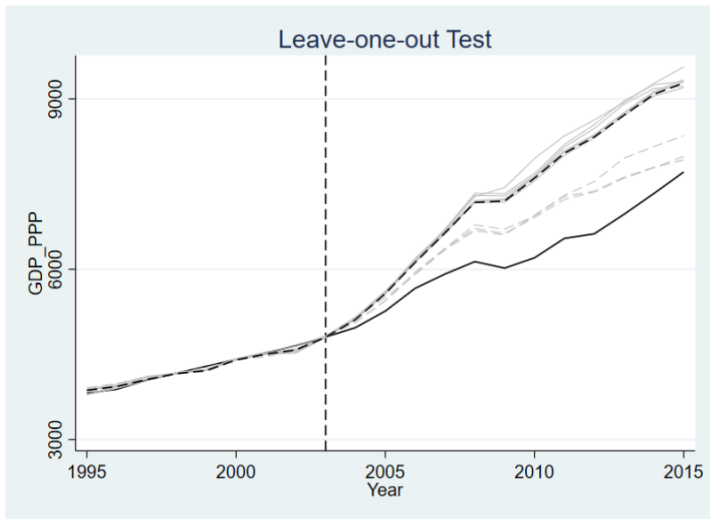


Figure 5.5: Treatment period in 2001

Robustness Tests in Synth Control

- **Leave-One-Out Test** → tests the sensitivity of the estimates, removing in each iteration one of the countries with positive weight in the donor pool.
Check if results are driven by any particular country
- **Reducing the number of units** in the synthetic control → allow the synthetic to be constructed only by the donors with positive weights. Then, iteratively, remove those receiving weight = 0 or the one with the smaller weight.

Examples



Can we do better?

- **Augmented SC** → outcome model to correct from poor fit bias
- **Synthetic DiD** → allows a level shifts, so weights need only to match trends, not levels

Augmented SC

Idea: clean the estimate from pre-treatment fit bias, using an outcome model.

ASCM estimator:

$$\hat{\theta}_{ASCM} = \underbrace{Y_{1,post} - \sum_j w_j^* Y_{j,post}}_{\text{Standard SC estimate}} - \underbrace{\hat{m}(X_1) + \sum_j w_j^* \hat{m}(X_j)}_{\text{Bias Correction}}$$

where $\hat{m}(\cdot)$ is an outcome model fit on the donor pool.

In the original paper a *Ridge outcome model* is used. Ridge is chosen because it can extrapolate beyond the convex hull in a regularized way, and because the resulting estimator has a clean equivalent representation as a synthetic-control estimator with modified weights, potentially including negative weights. Check **Ben-Michael et al. (2021)** for more.

Idea: create a synthetic control unit that is similar to treated only in trends (ie. similar up to an intercept, PTA that holds by construction). Then, run a classic DiD.

Formally, SDiD estimator solves a weighted TWFE problem of the form:

$$(\hat{\tau}, \hat{\alpha}, \hat{\beta}) = \operatorname{argmin}_{\tau, \alpha, \beta} \sum_{i,t} \hat{\omega}_i \hat{\lambda}_t (Y_{it} - \alpha_i - \beta_t - W_{it}\tau)^2$$

where:

- ω_i are unit weights as classic SC
- λ_t are time weights $\rightarrow$ give more weight to pre-treatment periods that are most informative about the untreated post-treatment trends

So in SDiD you have together: unit FE, time FE, unit weights, time weights.

First step into time weights

Time weights λ are chosen to solve a balancing problem over control units:

$$(\hat{\lambda}_0, \hat{\lambda}) = \operatorname{argmin}_{\lambda_0, \lambda \in \Lambda} \sum_{i=1}^{N_{con}} \left(\lambda_0 + \sum_{t=1}^{T_{pre}} \lambda_t Y_{it} - \frac{1}{T_{post}} \sum_{t=T_{pre}+1}^T Y_{it} \right)^2$$

SDID chooses the pre-period weights so that, for untreated units, the weighted average of pre-treatment outcomes resembles the average post-treatment outcome up to a constant

Check for more in **Arkhangelsky et al.** if interested